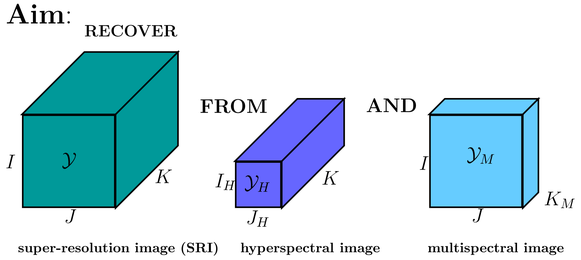
Coupled tensor low-rank multilinear approximation for hyperspectral super-resolution

Clémence Prévost[1], Konstantin Usevich[1], Pierre Comon[2], David Brie[1]

[1] CRAN, Université de Lorraine, CNRS, Vandoeuvre-lès-Nancy, France [2] GIPSA-Lab, Univ. Grenoble Alpes, CNRS, Saint-Martin d'Hères, France

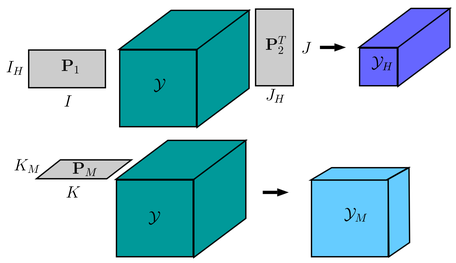
Hyperspectral super-resolution

Aim:



Degradation model:

$$\begin{cases} \mathcal{Y}_H &= \mathcal{Y} \bullet_1 \boldsymbol{P}_1 \bullet_2 \boldsymbol{P}_2 + \mathcal{E}_H, \\ \mathcal{Y}_M &= \mathcal{Y} \bullet_3 \boldsymbol{P}_M + \mathcal{E}_M \end{cases}$$



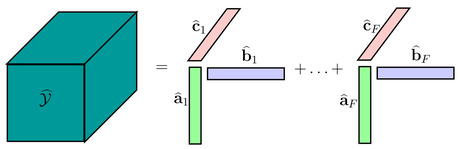
Low-rank approximation

Main idea: minimize $f(\widehat{\mathcal{Y}})$ where

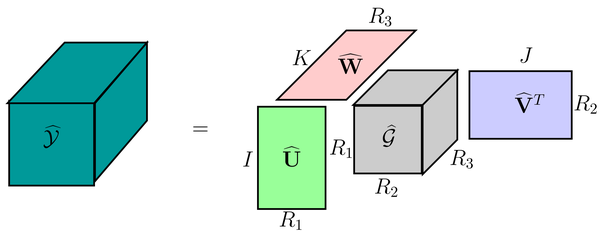
$$f(\widehat{\mathcal{Y}}) = \|\mathcal{Y}_H - \widehat{\mathcal{Y}} \underset{1}{\bullet} \boldsymbol{P}_1 \underset{2}{\bullet} \boldsymbol{P}_2\|_F^2 + \lambda \|\mathcal{Y}_M - \widehat{\mathcal{Y}} \underset{3}{\bullet} \boldsymbol{P}_M\|_F^2,$$

and $\widehat{\mathcal{Y}}$ is a low-rank tensor.

STEREO [2]: CP decomposition



This paper: Tucker decomposition



$$= [\![\widehat{\mathcal{G}}, \widehat{\mathbf{U}}, \widehat{\mathbf{V}}, \widehat{\mathbf{W}}]\!]$$

SCOTT: non-blind algorithm

input : $\mathcal{Y}_M, \mathcal{Y}_H, \boldsymbol{P}_M, \boldsymbol{P}_1, \boldsymbol{P}_2; [R_1, R_2, R_3]$
output: $\widehat{\mathcal{Y}}$

1. $\widehat{\boldsymbol{U}} \leftarrow \mathrm{tSVD}_{R_1}\left(\boldsymbol{Y}_M^{(1)}\right), \widehat{\boldsymbol{V}} \leftarrow \mathrm{tSVD}_{R_2}\left(\boldsymbol{Y}_M^{(2)}\right),$
$\widehat{\boldsymbol{W}} \leftarrow \mathrm{tSVD}_{R_3}\left(\boldsymbol{Y}_H^{(3)}\right)$
2. $\widehat{\mathcal{G}} \leftarrow \underbrace{\underset{\mathcal{G}}{\operatorname{argmin}}\ f\left([\![\mathcal{G}; \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}}]\!]\right)}_{\text{least squares problem}}$
3. $\widehat{\mathcal{Y}} = [\![\widehat{\mathcal{G}}; \widehat{\boldsymbol{U}}, \widehat{\boldsymbol{V}}, \widehat{\boldsymbol{W}}]\!]$

$\boldsymbol{Y}^{(i)}$ denotes the i-th standard unfolding of $\mathcal{Y}$.

Example - Indian Pines dataset

Algorithm	R-SNR	CC	SAM	ERGAS	time
STEREO 100	28.56	0.92	2.01	0.87	2.1
SCOTT [40,40,6]	26.39	0.89	2.32	1.06	0.18
SCOTT [30,30,16]	25.15	0.87	2.5	1.18	0.37
SCOTT [70,70,6]	27.62	0.9	2.19	0.95	0.33
SCOTT [24,24,25]	25.08	0.88	2.44	1.17	0.15
HySure E = 16	28.3	0.92	2.09	0.87	16.06

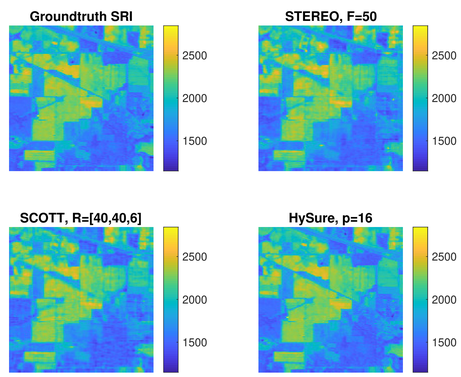
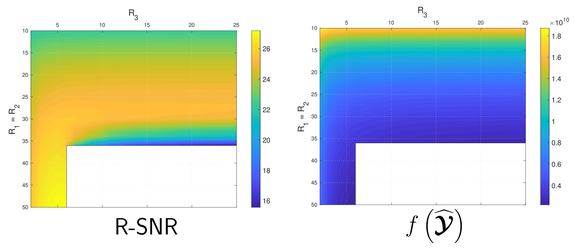


Figure 1:Spectral slice 120 of the SRI, Indian Pines



$$\text{R-SNR} = 10\log_{10}\left(\frac{\|\mathcal{Y}\|_F^2}{\|\widehat{\mathcal{Y}} - \mathcal{Y}\|_F^2}\right)$$

Recoverability of the model

Theorem ([5], deterministic recoverability)
Let $\mathcal{Y} = [\![\mathcal{G}; \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{W}]\!]$ be a Tucker decomposition of m.l. rank $[R_1, R_2, R_3]$ and $\mathcal{E}_M, \mathcal{E}_H = \mathbf{0}$.

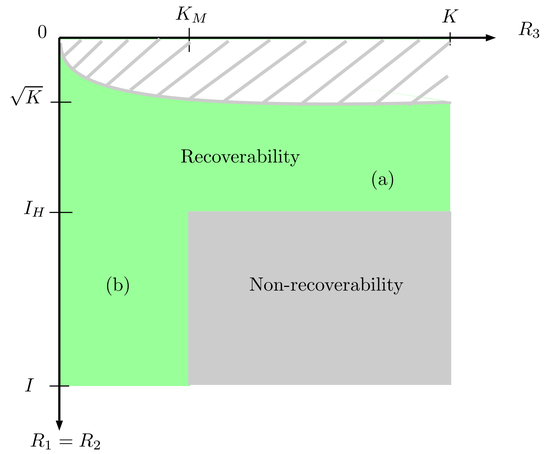
1. If $\mathrm{rank}\{\boldsymbol{Y}_M^{(1)}\} = R_1$, $\mathrm{rank}\{\boldsymbol{Y}_M^{(2)}\} = R_2$, $\mathrm{rank}\{\boldsymbol{Y}_H^{(3)}\} = R_3$ and one of the conditions holds:
   a) $\mathrm{rank}\{\boldsymbol{P}_1\boldsymbol{U}\} = R_1$ and $\mathrm{rank}\{\boldsymbol{P}_2\boldsymbol{V}\} = R_2$;
   b) $\mathrm{rank}\{\boldsymbol{P}_M\boldsymbol{W}\} = R_3$.
   Then there is only one $\widehat{\mathcal{Y}}$ of m.l. rank $[R_1, R_2, R_3]$ such that $\widehat{\mathcal{Y}} \bullet_3 \boldsymbol{P}_M = \mathcal{Y}_M$ and $\widehat{\mathcal{Y}} \bullet_1 \boldsymbol{P}_1 \bullet_2 \boldsymbol{P}_2 = \mathcal{Y}_H$.
2. If a) and b) are not satisfied, then there exist infinitely many such $\widehat{\mathcal{Y}}$.

Generic recoverability: Assume that

- $\boldsymbol{P}_1, \boldsymbol{P}_2, \boldsymbol{P}_M$ are fixed full row-rank, $\mathcal{E}_M, \mathcal{E}_H = \mathbf{0}$;
- $\mathcal{G}$, $\boldsymbol{U}$, $\boldsymbol{V}$ and $\boldsymbol{W}$ are random;
- $R_1 = R_2$, $I_H = J_H$ and $I = J$.

$$\left.\begin{array}{l} R_3 \leq \min(R_1, I_H)^2, \\ a) R_1 \leq I_H \text{ or } b) R_3 \leq K_M \end{array}\right\} \Rightarrow \text{unique recovery}$$

a) and b) not satisfied $\Rightarrow$ non-unique recovery



B-SCOTT: blind algorithm

input : $\mathcal{Y}_M, \mathcal{Y}_H, \boldsymbol{P}_M; [R_1, R_2, R_3]$
output: $\widehat{\mathcal{Y}}$
Split $\mathcal{Y}_M$, $\mathcal{Y}_H$ in L spatial patches.
for l=1:L **do**

1. $[\![\widehat{\mathcal{G}}_l; \widehat{\boldsymbol{U}}_l, \widehat{\boldsymbol{V}}_l, \widehat{\boldsymbol{W}}_l]\!] \overset{\text{HOSVD}}{\approx} \mathcal{Y}_{M,l}$
2. $\boldsymbol{Z}_l \leftarrow \mathrm{tSVD}_{R_3}\left(\boldsymbol{Y}_{H,l}^{(3)}\right)$
3. $\widehat{\boldsymbol{W}}_l \leftarrow \boldsymbol{Z}_l(\boldsymbol{P}_M\boldsymbol{Z}_l)^{\dagger}\widehat{\boldsymbol{W}}_l$
4. $\widehat{\mathcal{Y}}_l = [\![\widehat{\mathcal{G}}_l; \widehat{\boldsymbol{U}}_l, \widehat{\boldsymbol{V}}_l, \widehat{\boldsymbol{W}}_l]\!]$

end

Example - Pavia dataset

Algorithm	R-SNR	CC	SAM	ERGAS	time
SCUBA (4,4) [120,3]	25.67	0.99	3.25	1.97	17.6
B-SCOTT (4,4) [152,84,3]	26.42	0.99	2.97	1.83	0.55
SCUBA (8,8) [120,3]	26.49	0.99	2.93	1.81	49.35
B-SCOTT (8,8) [76,42,3]	26.52	0.99	2.92	1.84	0.73
Blind-STEREO F = 300	23.14	0.98	3.96	2.45	66.97
HySure E = 9	26.05	0.99	2.9	1.79	101.54

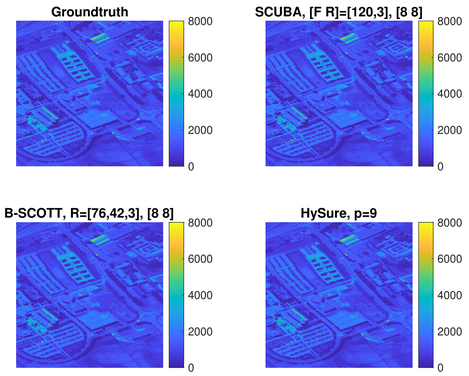


Figure 2:Spectral slice 44 of the SRI, Pavia University

[1] N. Yokoya, C. Grohnfeldt and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature". In *IEEE Trans. Geosci. Remote Sens.*, vol. 5, no. 2, pp. 29–56, 2017.

[2] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos and W.-K. Ma, "Hyperspectral Super-Resolution: A Coupled Tensor Factorization Approach". In *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6503–6517, 2018.

[3] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos and W.-K. Ma, "Hyperspectral Super-Resolution: Combining Low Rank Tensor and Matrix Structure". In *25th IEEE International Conf. on Image Process. (ICIP)*, Oct. 2018.

[4] C. Prévost, K. Usevich, P. Comon, and D. Brie, "Coupled Tensor Low-rank Multilinear Approximation for Hyperspectral Super-resolution". In *ICASSP*, Brighton, UK, 2019.

[5] C. Prévost, K. Usevich, P. Comon, and D. Brie, "Hyperspectral Super-Resolution with Coupled Tucker Approximation: Recoverability and SVD-based algorithms". https://hal.archives-ouvertes.fr/hal-01911969.

[6] http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

This work was partially supported by ANR grant OPTIFIN (ANR-15-CE10-0007).